



“He had engraved on it a map of the seven climates, and their lands and regions.”

Muhammad al-Idrisi, *Nuzhat al-Mushtaq fi Ikhtiraq al-Afaq*
(Book of Pleasant Journeys into Faraway Lands), 1154



Illustration of a water pump designed by al-Jazari

1206

Mechanical devices

In *The Book of Knowledge of Ingenious and Mechanical Devices*, Arab engineer al-Jazari described more than 50 machines and gave instructions for building them.

Among them were the first crankshaft (to convert circular motion into back-and-forth motion) and a 6.5-ft- (2-m-) high water clock in the shape of an elephant.

1237

Major medical book for women

Chinese physician Chen Ziming wrote *The Great Treatise of Beneficial Formulae for Women*, the first major Chinese medical work on treating women. It described 360 female medical conditions, as well as problems linked to pregnancy and childbirth.

1205

1225

1245

Healing people
See pages
76–77



1214

Use of antiseptics in Italy

Hugh of Lucca, an Italian surgeon, described how wine could be used as an antiseptic to clean wounds and prevent infection. Traditionally, doctors had thought, wrongly, that letting pus form in a wound helped injuries to heal.

Robert Grosseteste was the first chancellor of the University of Oxford, in England, from 1214.



1225

A bishop's theories

Robert Grosseteste, Bishop of Lincoln in England, tried to show how the philosophy and science of the Ancient Greek Aristotle (see pp.30–31) agreed with Christian ideas. Grosseteste held the belief that light fills the Universe and shapes its form. He thought scientific theories were best examined through experiments and that ideas not supported by observation should be rejected.

13th-century portrait of Bishop Grosseteste



1232

Gunpowder rockets

The first military use of rockets propelled by gunpowder was made by the Chinese against the Mongols during their siege of the town of Kaifeng in north-eastern China. These “flying-fire arrows” consisted of bamboo tubes filled with gunpowder attached to a stick. They were very inaccurate, but still caused the Mongols to abandon the siege and flee.